Refined Research Objectives

Open University Learning Analytics Dataset (OULAD)
Five Q1-oriented study proposals, based on a literature scan and first-pass data exploration
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1. Purpose

This document explains the research gaps found in the OULAD literature and refines an initial list of 16 candidate objectives into five study proposals. Each one addresses a gap in the existing OULAD literature and can support a stand-alone journal article. Each objective states the real-world problem, the gap, the evidence already seen in the data, research questions, data and methods, evaluation metrics, expected contribution and risks.

2. Dataset at a glance

Table	Rows	Contents
studentInfo	32,593	Demographics, prior attempts, credits, final_result (target)
studentRegistration	32,593	Registration and unregistration day
studentAssessment	173,912	Submission day, score, banked flag
assessments	206	Type (TMA / CMA / Exam), due date, weight
studentVle	10,655,280	Daily clicks per student per VLE resource
vle	6,364	Resource activity type and planned weeks
courses	22	Module-presentation length

Source: Kuzilek, Hlosta and Zdrahal (2017), *Scientific Data* 4:170171; CC-BY 4.0; obtained from the UCI ML Repository (id 349).

3. Key observations from Step 01 (studentInfo)

- Outcome distribution: Pass 37.9%, Withdrawn 31.2%, Fail 21.6%, Distinction 9.3%. Withdrawal is the dominant problem.
- 28,785 unique students over 32,593 enrolments: 3,808 repeat enrolments (a leakage risk).
- Only imd_band has missing values (3.41%); one band label is inconsistent ('10-20').
- Small groups: age 55+ (0.66%), no formal qualifications (1.06%), postgraduate (0.96%).
- Right-skewed numeric attributes: num_of_prev_attempts (skew 3.81), studied_credits (skew 1.88, max 655).

4. What the literature has already covered

- **Saturated:** accuracy comparisons of ML / DL dropout models (AUC up to 0.91 reported in 2019); adding SHAP / LIME explanations to a classifier.
- **Active but unresolved:** fairness (conflicting results), leakage-safe validation, assessment trajectories, cross-course generalisation.
- **Implication:** a new paper must offer a methodological or substantive insight, not a new accuracy score.

5. Research gaps explained

A **research gap** is a real-world question that earlier papers have not answered, have answered badly, or have answered with conflicting results. Each gap below states what it means, gives an example from OULAD, what has been done, and what is still missing.

Gap 1. Reported accuracy may be inflated by data leakage *(objective P1)*

What it means	*Data leakage* means the model sees information during training or testing that it would not have in real use, so the paper's result looks better than reality.
Example	(a) **Same student in train and test:** 32,593 rows but 28,785 students, so 3,808 repeat enrolments. A random split can put a student's first module in training and second module in test, and the model partly recognises the student, like grading an exam after seeing that student's earlier answer sheet. (b) **Future information:** a paper claims to predict at week 4 but uses a score released in week 6. OULAD does not record when scores were released, which makes this easy to get wrong.
Already done	Recent papers show that both problems inflate AUC and propose safe pipelines.
Still missing	No one has measured how much well-known published OULAD results are inflated, or whether the 'best model' is still best once the leakage is removed.

Gap 2. The assessment design itself is not studied *(objective P2)*

What it means	Almost every paper asks *which student* will drop out, using that student's behaviour. Very few ask whether *the way the course is assessed* pushes students out.
Example	Module X sets its first tutor-marked assignment (TMA) on day 20, worth 25% of the grade. Module Y starts with a 0% computer-marked quiz (CMA) and sets its first TMA on day 45. If withdrawals spike just before day 20 in Module X, that is a course-design problem, not a student problem.
Already done	Studies of submission behaviour (late-submission ratio, score trend) used to predict dropout.
Still missing	Design properties (type, weight, deadline spacing, first-assessment date) treated as explanatory factors across modules, and whether coursework predicts exam results equally in every module. Course teams can change deadlines far more easily than they can change students.

Gap 3. Fairness is checked one attribute at a time *(objective P3)*

What it means	A model is unfair if it makes more mistakes for one group than another, for example missing at-risk disabled students more often.
Example	A model can look fair for gender and fair for deprivation separately, yet badly miss **disabled students from deprived areas over 35**. Studying combined groups is called *intersectionality*. Also, only 216 students are aged 55+: an error rate of 30% vs 25% may be random noise without a confidence interval.
Already done	Single-attribute fairness audits; mitigation results that conflict (one study finds that it helps, another that it widens the gaps).
Still missing	An intersectional audit that reports uncertainty and claims only the gaps that are statistically reliable.

Gap 4. Missing data may itself be a warning sign *(objective P3)*

What it means	imd_band (deprivation of the home area) is missing for 1,111 rows (3.41%). Most papers delete those rows.
Example	Data is not always missing by chance: the address may not have matched or registration may have been incomplete. If these students withdraw more often, deleting them hides a vulnerable group, and the missingness could even help predict risk (MCAR vs MAR/MNAR).
Already done	Rows with missing values are usually dropped, or filled with the most common value.
Still missing	A test of whether the missingness is informative. **This is a hypothesis; it will be tested in Step 02** (e.g. chi-square of 'imd missing' vs final_result).

Gap 5. Accuracy alone is misleading ***(objective All)***

What it means	Accuracy hides class imbalance and says nothing about whether risk probabilities can be trusted.
Example	Only 9.3% of rows are Distinction, so a model that never predicts Distinction is still 90.7% accurate on that class. *Calibration:* if a model gives 100 students a 70% risk, about 70 of them should actually withdraw.
Already done	Most OULAD papers report accuracy, F1 or AUC.
Still missing	Routine reporting of PR-AUC, Brier score / calibration plots and confidence intervals.

Gap 6. When to intervene, given limited staff ***(objective P4)***

What it means	Papers predict one end-of-course label. Universities need to know which students to contact each week, and from which week the model can be trusted.
Example	Advisors can call 50 students a week. Predicting in week 2 leaves time to help but is less accurate; predicting in week 10 is more accurate but many students have already left.
Already done	Single-label prediction; an emerging survival benchmark (arXiv 2604.08870).
Still missing	Survival analysis (*when* withdrawal happens) linked to real advisor capacity, using precision@k. OULAD records the exact unregistration day, which makes this possible.

Gap 7. Explanations may not transfer to new courses ***(objective P5)***

What it means	Models trained on old courses are reused on new ones. Both their accuracy and their explanations may change.
Example	SHAP says a student was flagged for 'low forum activity'. That is meaningful in a discussion-based module and meaningless in a module without a forum. Train on 6 modules and test on the 7th, or train on 2013 and test on 2014.
Already done	SHAP applied on a single train/test split.
Still missing	Peer-reviewed tests of explanation stability across unseen modules and presentations. Note: SHAP shows what the model uses, not what causes withdrawal.

#	Gap in one sentence	Objective
1	Accuracy inflated by the same student or future data leaking into training	P1
2	Assessment design (type, weight, deadlines) not tested as a driver of withdrawal	P2
3	Fairness checked one attribute at a time, with no uncertainty for small groups	P3
4	Missing deprivation data may mark a hidden vulnerable group	P3
5	Accuracy hides imbalance and untrustworthy probabilities	All
6	No answer to when to intervene with limited advisors	P4
7	Explanations may change when a model is reused on a new course	P5

P1. Leakage-aware re-evaluation of OULAD early-warning models

Real-world problem

Universities deploy early-warning systems trusting published accuracy figures. If those figures are inflated by flawed evaluation, institutions over-trust the model and misallocate support staff.

Gap in the literature

Recent work (Information 17(6):581; arXiv 2605.25794) shows that row-level splits and post-cutoff assessment features inflate AUC. A regional fairness audit (Sustainability 18:9440) flags the same issue. However, **no study systematically quantifies how much published OULAD results are inflated**, or whether model rankings change.

Evidence from our data

32,593 enrolments but only 28,785 unique students, so 3,808 repeat enrolments can fall on both sides of a random split.

Research questions

- RQ1. How much do AUC / F1 change between a row-level random split and a student-grouped, time-ordered split?
- RQ2. How much of the apparent early-week performance comes from features computed after the prediction cutoff (e.g. assessment scores released later)?
- RQ3. Does the ranking of common models (LR, RF, XGBoost, LSTM) stay the same under leakage-safe evaluation?

Data tables	studentInfo, studentRegistration, studentAssessment, assessments, studentVle, vle
Method	Recreate 3-4 typical published pipelines; evaluate each under (a) a random row split, (b) GroupKFold by id_student, (c) a train-on-2013 / test-on-2014 presentation split, and (d) strict cutoff-first feature truncation at weeks 2, 4, 8 and 12.
Metrics	ROC-AUC, PR-AUC, macro-F1, Brier score; inflation = metric(leaky) - metric(safe) with bootstrap 95% CIs; Kendall's tau for rank stability.
Contribution	A quantified leakage audit and a reusable leakage-safe evaluation protocol for OULAD that other researchers can follow.
Key Pandas / NumPy	merge, groupby().agg, cutoff filtering with boolean masks, np.random.default_rng for reproducible bootstraps.
Risks / limits	Needs the heaviest ML work. Results depend on which published pipelines are chosen, so the selection must be justified.

P2. Assessment design and student outcomes in online distance education

Real-world problem

Course teams choose assessment types, weights and deadlines with little evidence. Poorly spaced or heavily weighted early assessments may push students to withdraw.

Gap in the literature

OULAD assessment studies (e.g. MDPI Data 11(11):281, 2026) use student-level submission behaviour (lateness, score slope) to **predict dropout**. The **design of the assessment model itself** (TMA vs CMA vs exam, weighting, timing, spacing) as an explanatory factor is largely unstudied.

Evidence from our data

206 assessments across 22 module-presentations, with type (TMA / CMA / Exam), weight and due date available for every assessment.

Research questions

- RQ1. How do assessment designs differ across modules (number, type mix, weighting, deadline spacing, date of the first assessment)?
- RQ2. Are design features associated with withdrawal and pass rates after controlling for student demographics and prior attempts?
- RQ3. Do coursework scores (TMA, CMA) predict exam performance equally well, and does that relationship vary by module?
- RQ4. Is withdrawal concentrated around specific assessment deadlines?

Data tables	assessments, studentAssessment, courses, studentInfo, studentRegistration
Method	Build presentation-level design descriptors; use multilevel (mixed-effects) logistic regression with students nested in presentations; deadline-aligned withdrawal event plots; correlations between TMA/CMA and exam scores with Fisher z comparisons.
Metrics	Odds ratios with 95% CIs, intraclass correlation (ICC), Spearman rho, effect sizes (Cramer's V, Cohen's d).
Contribution	Evidence-based guidance for designing assessment in online courses. Framed by assessment-for-learning theory and Tinto's model of student departure.
Key Pandas / NumPy	merge, pivot_table, groupby, np.diff for deadline spacing, pd.cut, crosstab, corr(method='spearman').
Risks / limits	Only 22 presentations exist at the design level, so the analysis must be framed as associational and the multilevel structure handled carefully.

P3. Intersectional fairness and informative missingness in at-risk prediction

Real-world problem

Early-warning tools may systematically under-flag or over-flag disadvantaged students. Missing socio-economic data may itself mark a vulnerable group that is silently dropped.

Gap in the literature

OULAD fairness studies (JLA; Frontiers in Education 2026; JEDM; MDPI Electronics 14:1856) examine **single attributes** and reach **conflicting conclusions** about mitigation. Intersectional groups, uncertainty for small groups, and whether IMD missingness is informative remain unstudied.

Evidence from our data

imd_band is 3.41% missing (1,111 rows) and has a label error ('10-20' without '%'); age 55+ is only 216 rows (0.66%); disability is 9.7%.

Research questions

- RQ1. Is missing imd_band associated with different outcomes (i.e. is it informative, not missing at random)?
- RQ2. Do error rates and calibration of an early-warning model differ across intersections (IMD x disability x age x education)?
- RQ3. Which gaps remain statistically reliable once uncertainty for small groups is accounted for?
- RQ4. Does group-aware recalibration reduce the gaps without reducing overall usefulness?

Data tables	studentInfo (core), plus early-week VLE and assessment features for the predictive model
Method	Missingness analysis (chi-square, logistic model of missingness); a leakage-safe baseline model; subgroup metrics with bootstrap CIs and Bayesian shrinkage for small groups; comparison of mitigations (reweighting, group-wise calibration).
Metrics	Equal-opportunity difference, FPR/FNR gaps, calibration error per group, Brier score, PR-AUC; 95% CIs throughout.
Contribution	The first intersectional, uncertainty-aware fairness audit of OULAD, and a test of whether missing data marks a disadvantaged group.
Key Pandas / NumPy	isnull, groupby on multiple keys, crosstab(normalize='index'), value_counts, np.percentile for bootstrap CIs.
Risks / limits	Intersections become very small, so the paper must report CIs and avoid over-claiming. OULAD has no ethnicity attribute.

P4. When to intervene: withdrawal timing and capacity-aware early warning

Real-world problem

Advisors can contact only a limited number of students each week. Institutions need to know **when** withdrawal risk peaks and how early a warning is accurate enough to act on.

Gap in the literature

Most OULAD models predict a single end-of-course label. Survival benchmarks are emerging (arXiv 2604.08870), but linking withdrawal timing to **intervention capacity** (e.g. the top-k students an advisor can call) is rare.

Evidence from our data

31.2% of enrolments end in withdrawal, the largest outcome problem in the data; studentRegistration records the exact unregistration day.

Research questions

- RQ1. What does the withdrawal hazard look like over the course weeks, and does it differ by module and student background?
- RQ2. From which week does a calibrated model give useful precision for the top-k flagged students?
- RQ3. How many withdrawals could be reached under realistic advisor capacities?

Data tables	studentRegistration, studentInfo, studentVle (weekly aggregates), studentAssessment, courses
Method	Kaplan-Meier curves and Cox / discrete-time hazard models; weekly landmark models with cutoff-first features; precision@k and capacity simulation.
Metrics	Concordance index, time-dependent AUC, Brier score, precision@k, recall@k.
Contribution	An operational answer to when and how many: guidance on intervention timing for online universities.
Key Pandas / NumPy	pd.cut into weeks, groupby + cumsum for weekly engagement, merge_asof-style cutoffs, np.where for event indicators.
Risks / limits	Survival analysis needs the lifelines package; withdrawals recorded before the course starts must be handled consistently.

P5. Cross-module transferability of at-risk predictors and their explanations

Real-world problem

A model trained on one course is often reused for a new course. If its predictors or explanations change, advisors receive misleading reasons for flagging a student.

Gap in the literature

SHAP is widely applied to OULAD models (i-JET; MDPI Algorithms 18:781), but few peer-reviewed studies test whether **explanations remain stable** on unseen modules or later presentations.

Evidence from our data

7 modules (AAA-GGG) of very different sizes (2.3% to 24.3% of enrolments) and 4 presentations, which is a natural setting for transfer tests.

Research questions

- RQ1. How much does predictive performance drop with leave-one-module-out and forward-in-time transfer?
- RQ2. Are the top SHAP features and their directions stable across modules?
- RQ3. Which features transfer reliably and which are module-specific?

Data tables	All tables; VLE activity types from vle
Method	Leave-one-module-out and 2013 to 2014 transfer; SHAP on tree models; rank-stability measures across folds.
Metrics	AUC / PR-AUC drop, Kendall's tau and Jaccard top-k overlap of feature rankings, sign agreement.
Contribution	Evidence on when an early-warning model, and its explanations, can safely be reused across courses.
Key Pandas / NumPy	pivot of activity_type clicks, groupby by module, rank(), set operations on top-k lists.
Risks / limits	SHAP explanations are not causal; the smallest module (AAA, 748 rows) gives unstable estimates.

6. Comparison of objectives

	Novelty	Effort	ML needed	Fits Pandas / NumPy focus	Main tables
P1 Leakage audit	High	High	Heavy	Medium	All
P2 Assessment design	High	Medium	Light	High	assessments, studentAssessment
P3 Intersectional fairness	Medium-high	Medium	Moderate	High	studentInfo (+ early features)
P4 Intervention timing	Medium	Medium-high	Moderate	Medium	studentRegistration, studentVle
P5 Cross-module transfer	Medium	High	Heavy	Medium	All

7. Recommendation

Start with **P2 (assessment design)** or **P3 (intersectional fairness)**. Both are less crowded, build on data already explored, and suit a statistics-led paper in which Pandas and NumPy carry most of the analysis. P1 offers the strongest methodological novelty and is a good second paper once the leakage-safe pipeline exists, because P3, P4 and P5 can reuse it.

8. Quality requirements for Q1 submission

- The contribution is a research finding about education; tool usage belongs in the Methods section.
- Theory framing: e.g. Tinto's model of student departure, self-regulated learning, assessment for learning.
- Leakage-safe validation: student-grouped and temporal splits; strict prediction cutoffs.
- Evaluation beyond accuracy: PR-AUC, calibration (Brier), subgroup metrics, 95% CIs, effect sizes.
- Actionable implications for institutions, and an honest limitations section (one UK university, 2013-2014 data, anonymised modules, no ethnicity or motivation data, clicks are proxies for learning).
- Reproducibility: public code, fixed random seeds, documented preprocessing.

Candidate venues (check the current quartile on SCImago / JCR before submitting): Computers & Education; British Journal of Educational Technology; The Internet and Higher Education; IEEE Transactions on Learning Technologies; Education and Information Technologies.

9. Key references (verify before citing)

[1] Kuzilek J., Hlosta M., Zdrahal Z. (2017). Open University Learning Analytics dataset. Scientific Data 4:170171.

[2] Jha N.I., Ghergulescu I., Moldovan A.-N. (2019). OULAD MOOC dropout and result prediction using ensemble, deep learning and regression techniques. CSEDU. doi:10.5220/0007767901540164

[3] Explainable early warnings for dropout through longitudinal assessment trajectories: evidence from OULAD (2026). Data (MDPI) 11(11):281.

[4] Time-consistent prediction in higher education: a framework for preventing data leakage in longitudinal models. Information (MDPI) 17(6):581. doi:10.3390/info17060581

[5] When can we trust early warnings? Leakage-excluded early outcome prediction from LMS interaction logs. arXiv:2605.25794.

[6] A unified survival benchmark for temporal dropout risk prediction in learning analytics. arXiv:2604.08870.

[7] Toward sustainable and equitable AI in education through a regional fairness audit of dropout prediction using OULAD. Sustainability 18:9440. doi:10.3390/su18189440

[8] Fairness mitigation on OULAD (2026). Frontiers in Education. doi:10.3389/feduc.2026.1841724

[9] Unveiling accuracy-fairness trade-offs: investigating ML models in student performance prediction. Journal of Learning Analytics.

[10] Investigating demographic features and their connection to performance, predictions and fairness in EDM models. Journal of Educational Data Mining.

[11] Evaluating fairness strategies in educational data mining. Electronics (MDPI) 14(9):1856.

[12] Early student risk detection using CR-NODE with explainable AI. Algorithms (MDPI) 18(12):781.

[13] Interpretable predictive modeling for educational equity: a workload-aware decision support system. Big Data and Cognitive Computing 9(11):297. doi:10.3390/bdcc9110297

[14] Predictive modelling with OULAD: a systematic literature review (2024); Alhakbani, Open learning analytics: a systematic review of benchmark studies using OULAD.

Full links are listed in literature/literature_review.md in the project repository.